United International University

School of Science and Engineering

Department of Computer Science and Engineering

Final Exam   Trimester:- Fall 2025

Course Title: Fundamental Calculus

Course Code: MATH-1151   Marks:- **40**   Time: **2** hours

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order.]

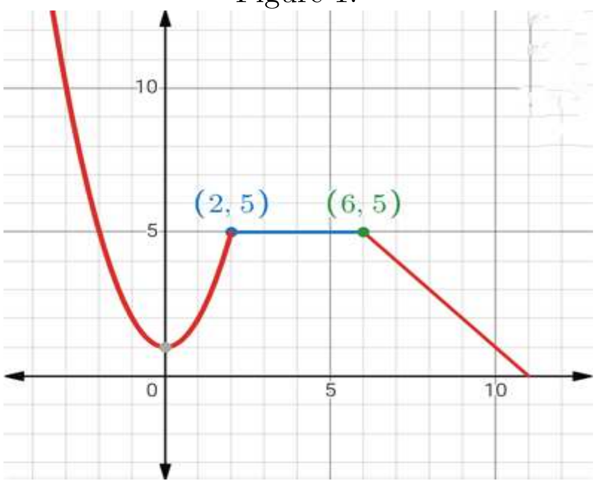
1. (a) Consider the function, $y = x^3 - 3x + 1$.

   i) Find the slope/gradient of the given function at $x = 2$. [2]

   ii) Find the equation of the tangent at $x = 2$. [2]

   (b) Draw the derivative curve function of the following function: [3]

Figure 1:



2. (a) The displacement of a particle is given by, $x = e^{2t} + \sqrt{t} + 10$

   i) Find the velocity at $t = 4$. [2]

   ii) Find the acceleration at $t = 1$. [2]

   (b) Find the coordinates of the turning point of the function,
   $y = x^3 - \frac{9}{2}x^2 - 12x - 8$. Also state their nature. [4]

   (c) Given that, $y = \sin 2x + e^{-2x} + x^2 - 1$. Find $\frac{d^2y}{dx^2}$. [2]

P.T.O.

3. (a) Given, $\frac{dy}{dt} = \frac{6}{\sqrt{3t-5}}$ and $y = 7$ when $t = 2$.
Find y in terms of t. [4]

(b) Differentiate the following with respect to x (Any two): [4]

i) $y = \ln\sqrt{\frac{x^2}{x^3-1}}$

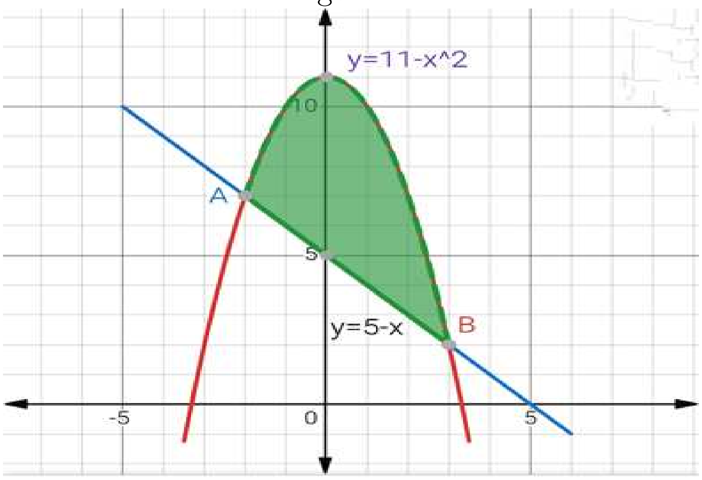
ii) $y = x\cos 3x + 7x$

iii) $y = \frac{e^x}{x^2+x}$

4. (a) Evaluate $\int_0^\pi t\sin(t)dt$. [3]

(b) The following figure shows the enclosed area between two curves $y = 11 - x^2$ and $y = 5 - x$.

Figure 2:



i) Find the x-coordinates of the points A and B. [1]

ii) Find the area of the shaded region. [3]

5. (a) Answer any two: [4]

i) $\int 2t^2\sqrt{3+t^3}dt$

ii) $\int_1^2 \frac{x^2+x+3}{x^2}dx$

iii) $\int \sin^4 x\cos x dx$

(b) Consider the function,

$$f(x) = \begin{cases} x+2, & -2 \le x < 2 \\ x^2, & 2 \le x \le 6 \end{cases}$$

Sketch the function and hence, evaluate $\int_{-2}^5 f(x)dx$. [4]